Owen Marzolf-Miller

###-###-#### | Contact Via Linkedin | College Park, MD - Willing To Relocate |

Education

University of Maryland — College Park, MD
Bachelor of Science in Computer Science: Machine Learning & Mathematics — GPA: 3.85 — May 2028
Lin. Alg., MV Calc., Number Theory, Probability, Algorithms, Compilers/Systems, Algorithmic Game Theory

National University of Singapore — Singapore
Exchange Coursework: Parallel Computing (CUDA, OpenMP), Effective Reasoning/Logic, Differential Equations — *Fall 2026*

Experience

Engineering Development Group Intern — May – August 2026
Statistics and Machine Learning Team at MathWorks — *Boston, MA*
- Researched & developed **partitioning algorithms** for structured datasets
- Tested top algorithms in empirical study, then implemented into **MATLAB**
- Led design & implementation of tabular data for **model feature selection**, unblocking customer workflows

Apex Quant @ UMD - Quantitative Analyst — September 2025 – Present
Volatility Strategies and Bond Options Market Research — *College Park, MD*
- Conducted research & backtested options trading strategies on several bond and SPY options markets
- Gathered 5+ years of historical data & implemented backtests in Python via **Pandas/NumPy** to evaluate spreads and volatility strategies (e.g. iron butterflies, iron condors) on the SPY options market
- Identified strategies & parameters to yield **1.58x** return on investment over an average of 7 trades per month

App Development Club @ UMD - iOS Engineer — September – December 2025
Financial App - Augmented Reality Frontend, Linear Regression Backend — *College Park, MD*
- Primary developer prototyping financial literacy app that teaches students budgeting and investment skills through interactive lessons and activities
- Built interactive-lessons in SwiftUI with AR; implemented **linear regression** to calculate future loans
- Delivered app to be demo-ed by UMD Dean for 50,000+ people during Maryland Day

Projects

Musical Tinder @ BitCamp — March 2025
JavaScript, Node.js, React, Python, Flask — *College Park, MD*
- Created full-stack app to swipe on new music suggestions and to add to Spotify/Apple Music playlists
- Designed the backend that connected app login to each user's Spotify & Apple Music; implemented it with 95% success for login connection and full playlist transfer

Web App for Alzheimer's Patients @ HackHarvard — October 2024
AWS Cloud, Python, JavaScript — *Boston, MA*
- Developed a web app that provides families with support for those with Alzheimer's through reminders, route notifications, and patient safety data
- Built the back-end that sent email notifications, kept patient data, and implemented login security for the web app through AWS cloud; connected front-end for patient login and to send 100% consistent family notifications

Medication Identifier @ DivHacks — October 2024
Python, TensorFlow, OpenCV — *New York City, NY*
- Designed a program to identify pill capsules via images for unlabeled bottles/pills
- Implemented identification software that classifies pills from computer/phone images through OpenCV and TensorFlow with 90% accuracy to help medication for unlabeled pills and visually impaired patients

Miscellaneous Projects | *C/C++, Assembly, OpenMP* — March 2025 - October 2026
- Developed and implemented a memory allocator system for data management in C and, again, in Assembly
- Optimized particle collision in physics simulator with OpenMP through SIMD & thread parallelization

Technical Skills

Languages: Python, C++, Swift, Assembly, OCaml, Java, Rust, JavaScript, HTML/CSS, MATLAB, SQL
Frameworks: Git, NumPy, PyTorch, Pandas, AWS, SwiftUI, OpenCV, React, Node.js, Flask, JUnit
Interests: Poker, Blackjack, NeurIPS, ICMT, Hackathons, Puzzles, Marathons/Triathlons